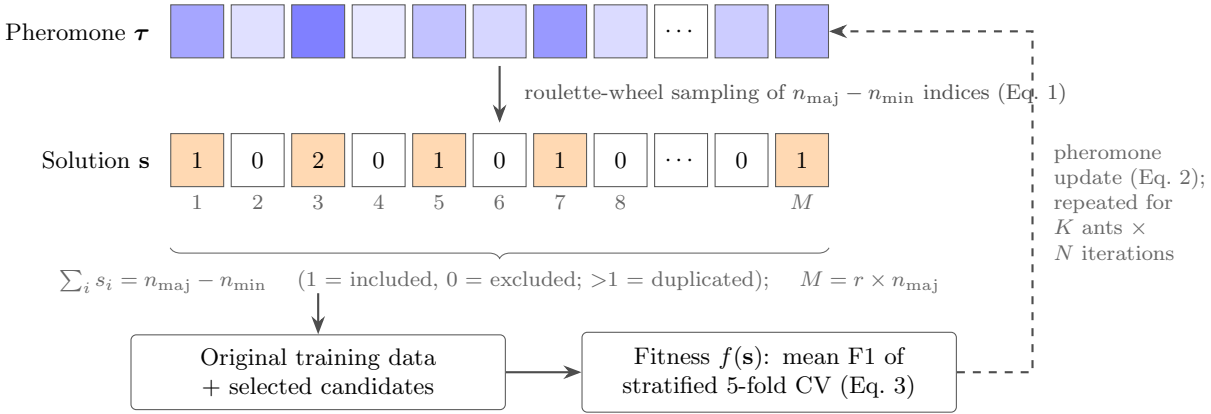

Pheromone $\boldsymbol{\tau}$
roulette-wheel sampling of $n_{\text{maj}} - n_{\text{min}}$ indices (Eq. 1)
Solution $\mathbf{s}$
1 0 2 0 1 0 1 0 ⋯ 0 1
1 2 3 4 5 6 7 8 M
pheromone update (Eq. 2); repeated for K ants × N iterations
$\sum_i s_i = n_{\text{maj}} - n_{\text{min}}$ (1 = included, 0 = excluded; >1 = duplicated); $M = r \times n_{\text{maj}}$
Original training data + selected candidates
Fitness $f(\mathbf{s})$: mean F1 of stratified 5-fold CV (Eq. 3)